

Ever wondered why the price of the same flight changes overnight?



Flight pricing is a complex challenge for everyone involved.



The Traveler's Dilemma

How can I find the best flight at the lowest possible price without constant monitoring?

- When is the best time to book?
- Which airline offers the best value?
- How much does an extra stop *really* cost?



The Airline's Challenge

How do we set competitive prices that attract customers while maximizing revenue?

- How should we adjust pricing for peak demand?
- What services are customers willing to pay more for?
- How do our prices compare to competitors?

Our goal was to build a tool that predicts flight fares and uncovers the key drivers behind their prices.



This project delivers two core outcomes:

1. **A Predictive Model:** An accurate machine learning model that estimates the cost of a flight based on its specific details.
2. **Actionable Insights:** A deep analysis of historical data to reveal patterns that can guide decision-making for both travelers and airlines.

The analysis is built on a dataset of Indian domestic flights with 11 distinct features.



Airline: The carrier operating the flight.



Date_of_Journey:
The day, month, and year of departure.



Source/Destination:
The departure and arrival airports.



Dep_Time/Arrival_Time:
The times of departure and arrival.



Duration: The total travel time.



Total_Stops: The number of layovers.



Additional_Info:
Details like baggage allowance.



Price: The target variable, in Indian Rupees.

We engineered new features from the raw data to better inform the model.

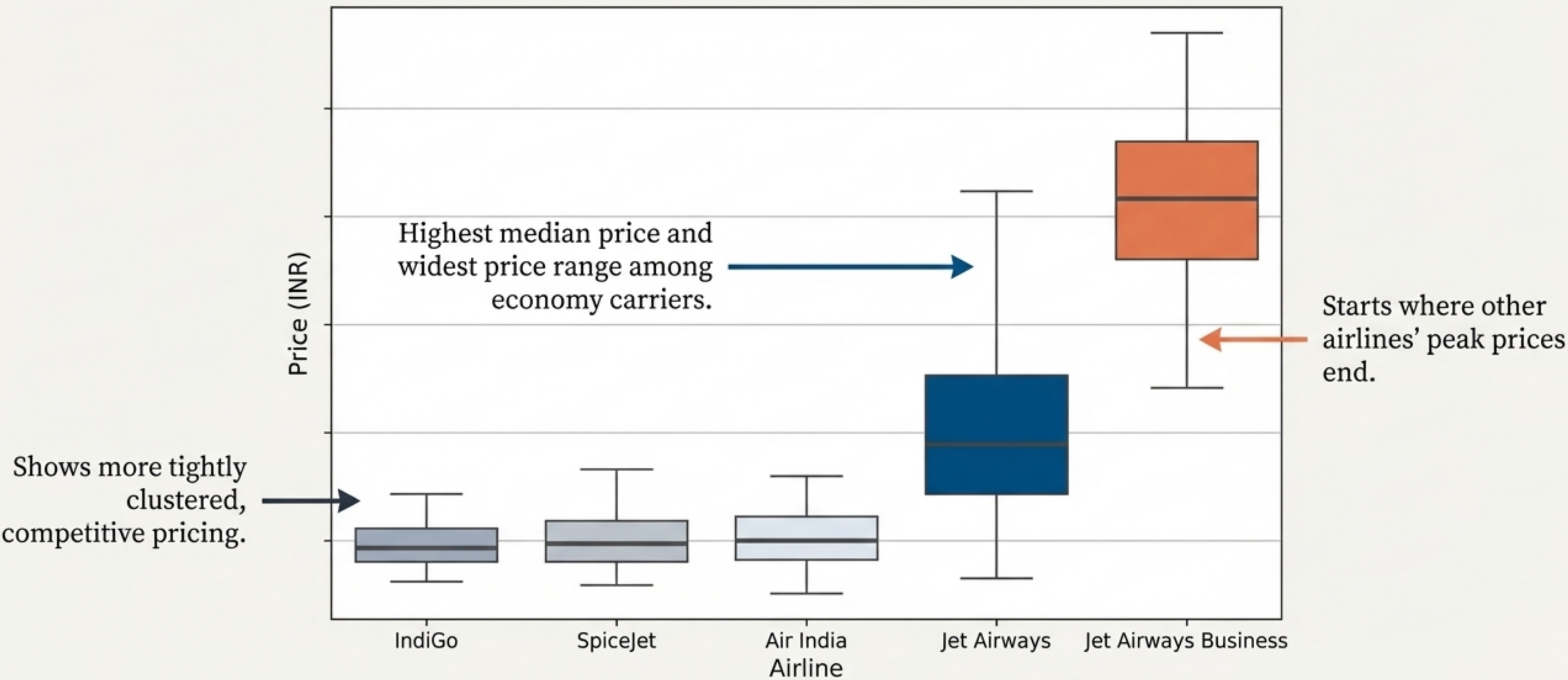


- **Data Cleaning:** Handled missing values (only two rows, which were removed).
- **Date & Time Extraction:** Deconstructed `Date\_of\_Journey` into `Day` and `Month` features. Similarly, `Hour` and `Minute` were extracted from departure and arrival times.
- **Feature Pruning:** The `Route` feature was dropped as it was redundant with `Total\_Stops`. `Additional\_Info` was also removed due to a lack of variance.
- **Categorical Encoding:** Used a hybrid strategy: One-Hot Encoding for features with fewer than 5 unique values and Label Encoding for those with more, balancing dimensionality and informational value.

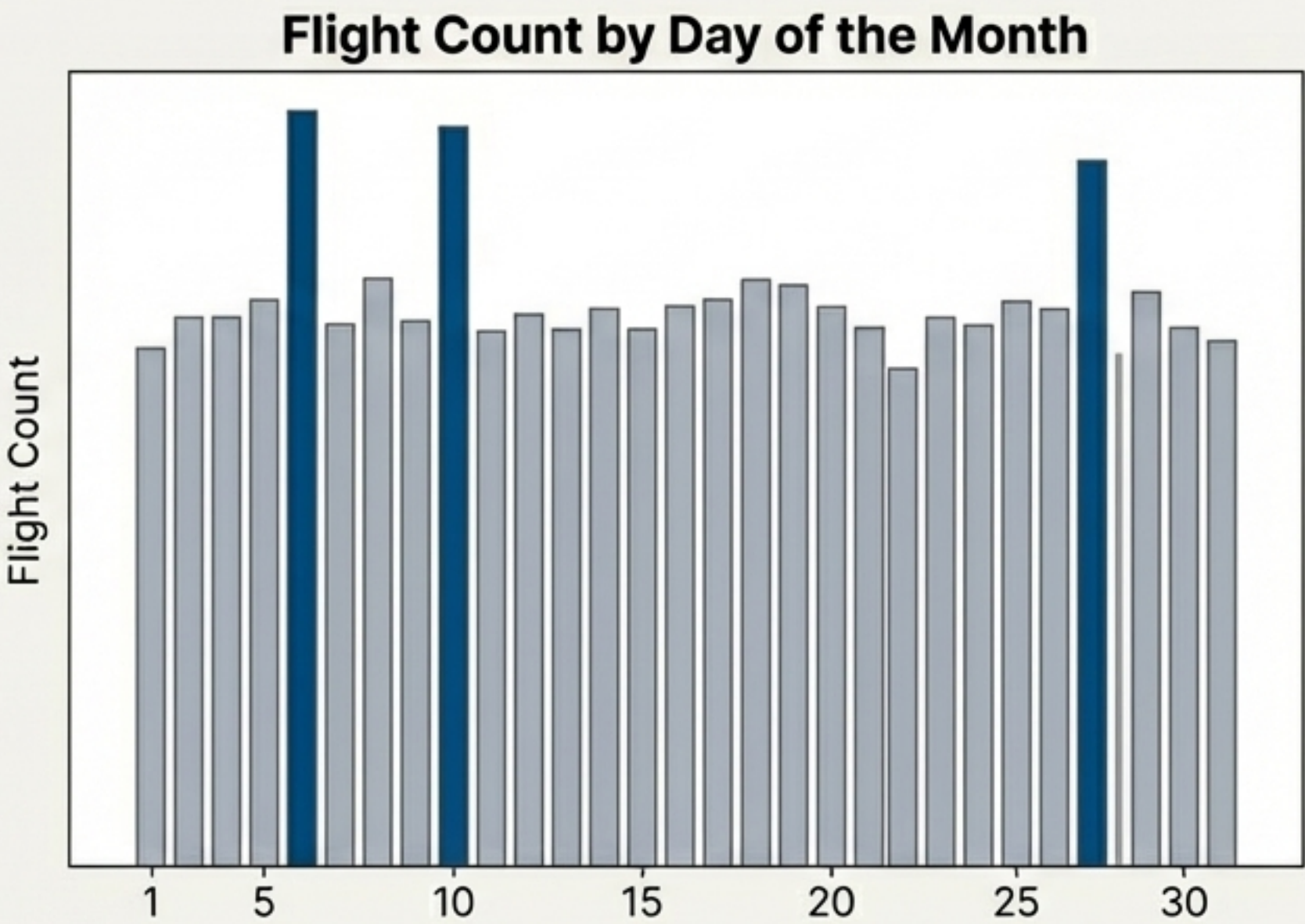
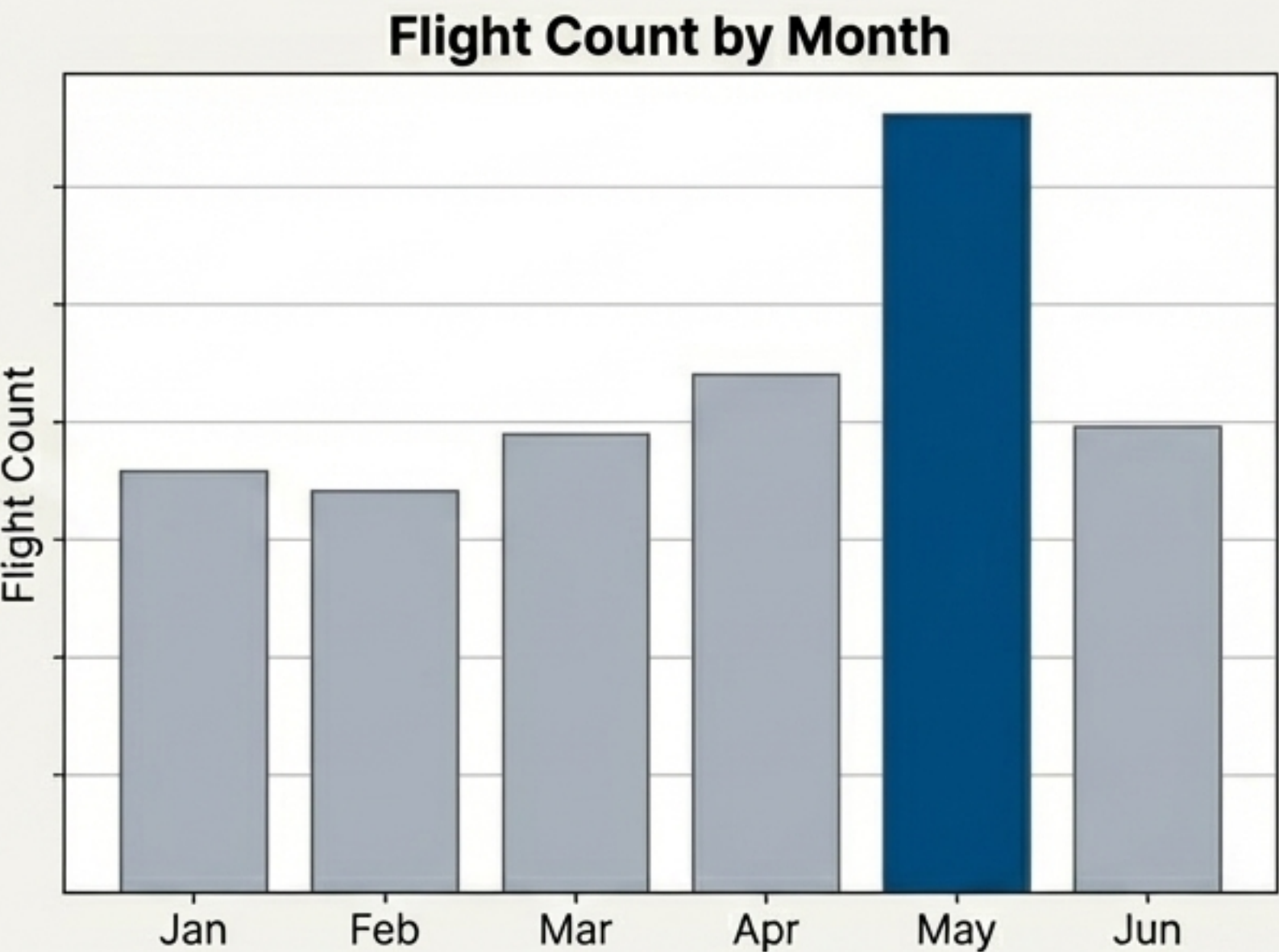
Insight: Direct flights aren't always the cheapest, but prices rise sharply after the first stop.



Insight: Jet Airways leads in flight volume and price, while its Business Class operates in a completely different price category.

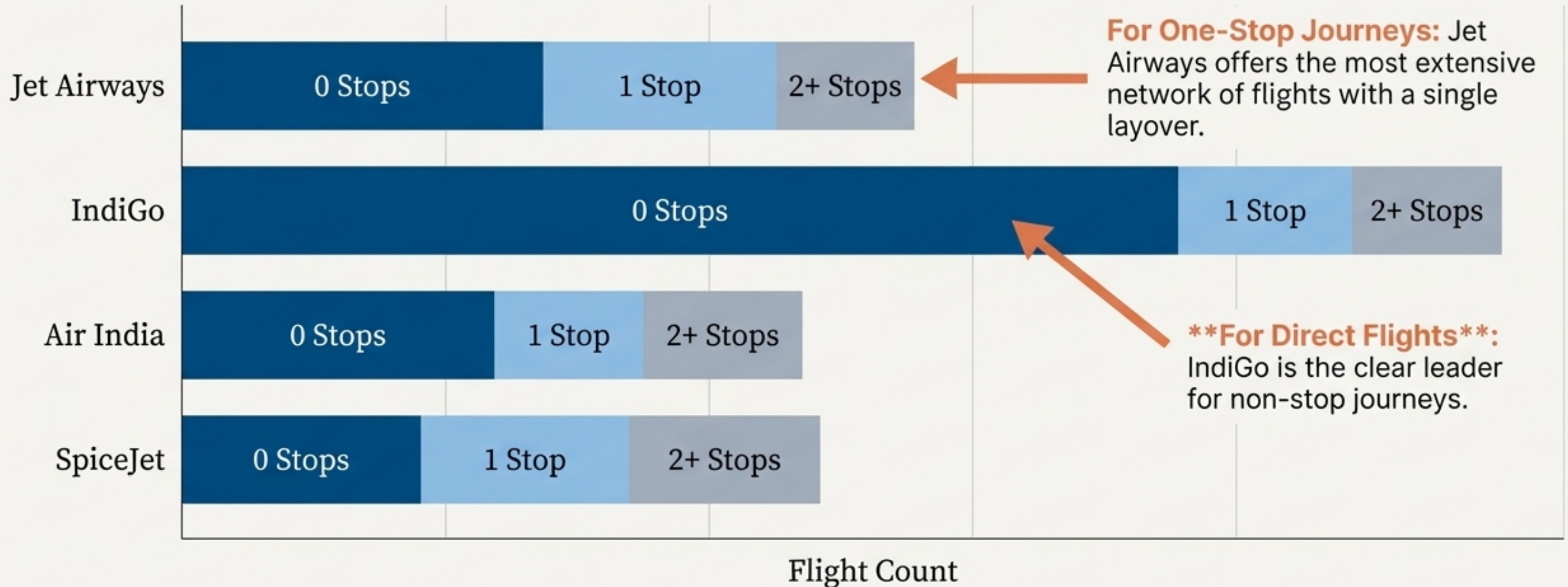


Insight: Peak travel demand clusters in early summer and on specific days of the month.

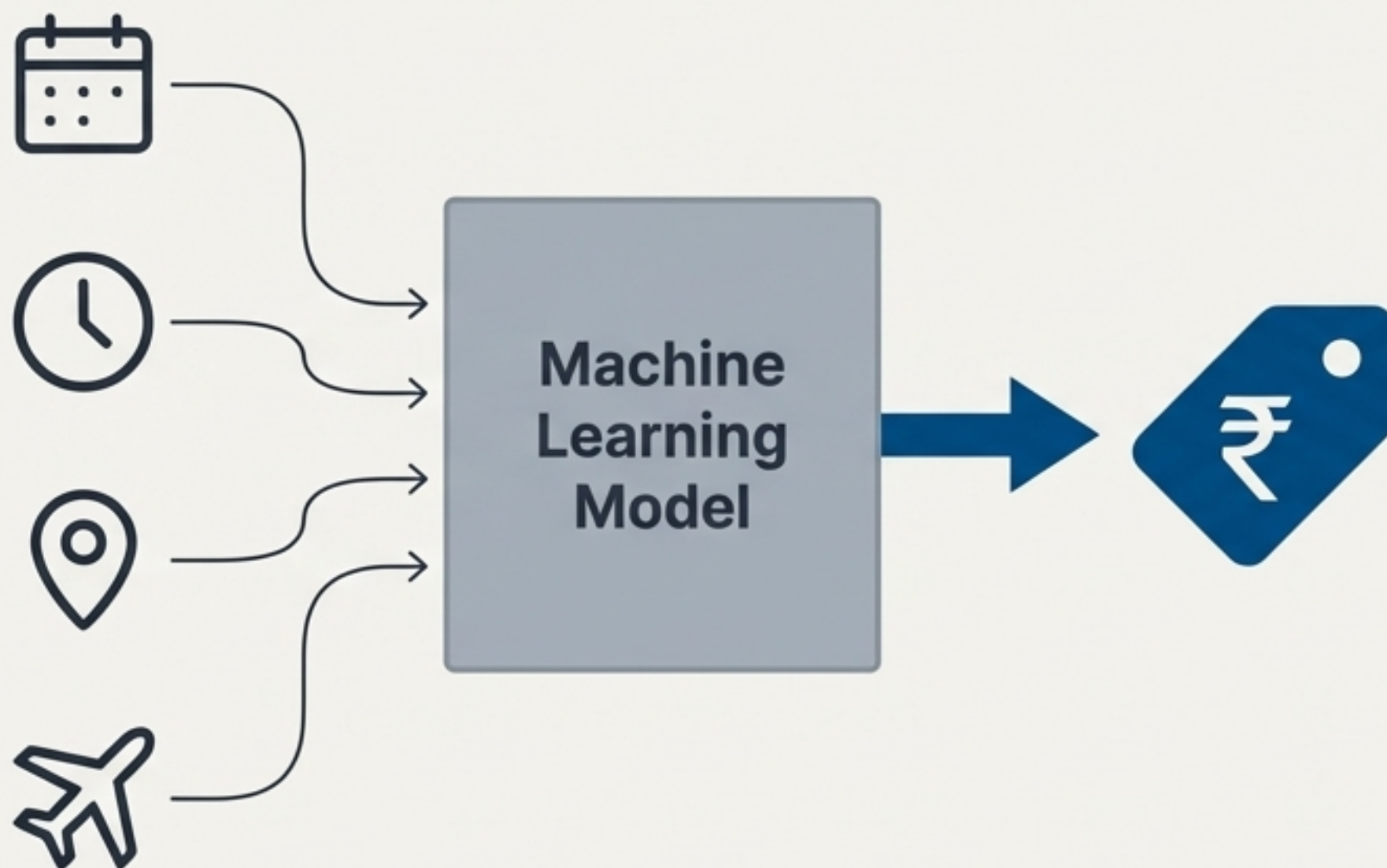


Airlines can strategically increase flight frequency during May and on these specific peak days to meet demand. Travelers looking for deals should aim for off-peak times.

Insight: Your choice of airline should depend on your priority: a direct flight or more options.



With a deep understanding of the data, we then built the predictive engine.



The Task

A regression problem to predict the continuous `Price` variable.

The Contenders

We evaluated a suite of robust regression algorithms to find the best performer for our specific dataset.

The Approach

A 'Model Bake-Off' where each algorithm is trained and tested under identical conditions to ensure a fair comparison. The contenders included Random Forest Regressor and XGBoost.

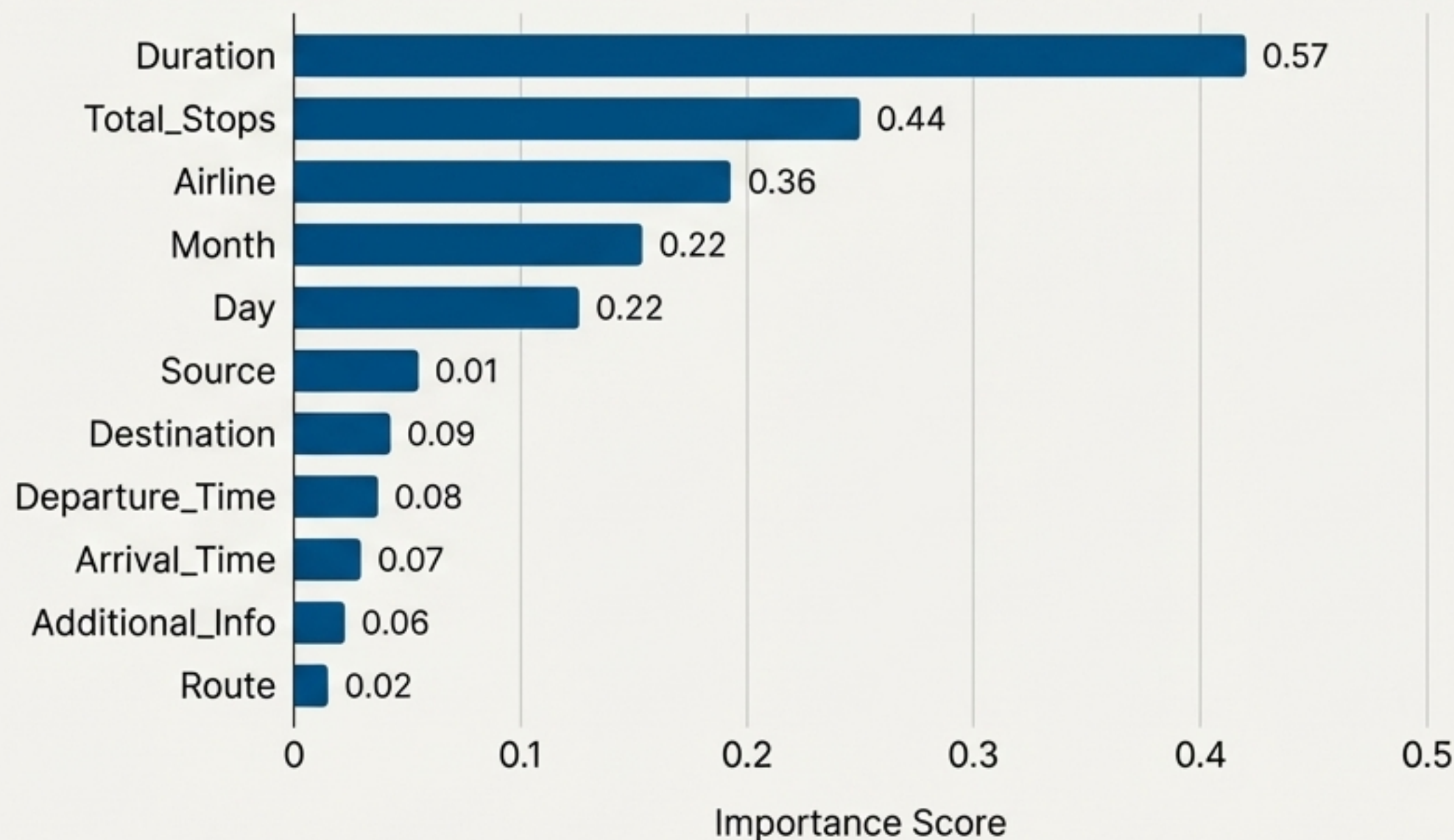
Random Forest Regressor emerged as the most accurate and reliable model.

Model	R ² Score	Root Mean Squared Error (RMSE)
Random Forest Regressor	0.99	458.32
XGBoost Regressor	0.98	512.19
[Other Models Tested]	...	...

Crucially, the chosen model's performance was validated on a completely separate, unseen dataset to ensure its real-world effectiveness, maintaining a high R² score of 0.97.

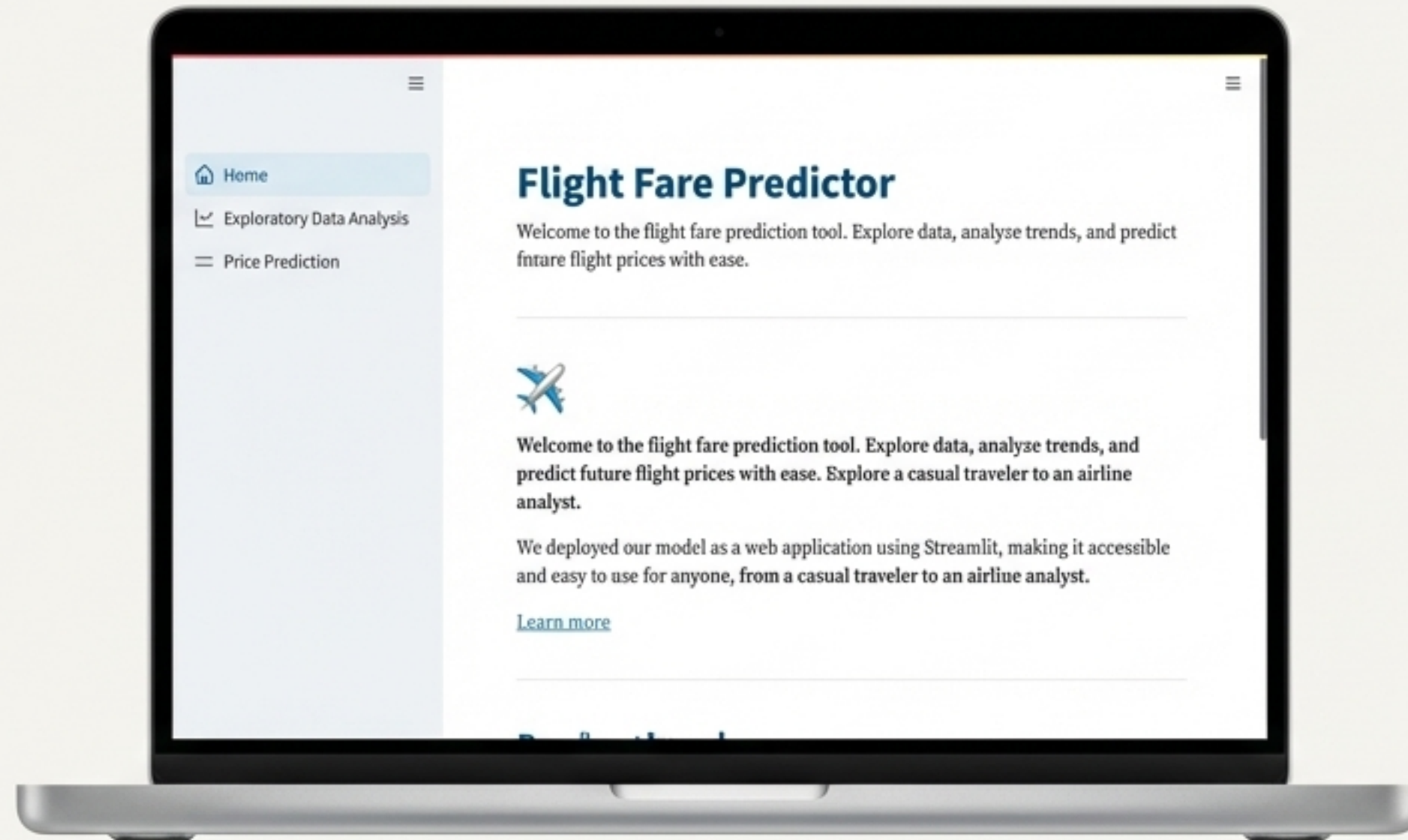
The model confirmed that flight duration, airline, and number of stops are the most significant price drivers.

Feature Importance Scores



This confirms our exploratory findings and gives us confidence that the model is learning logical and relevant patterns from the data.

The final result is not a report, but a fully interactive prediction tool.



We deployed our model as a web application using Streamlit, making it accessible and easy to use for anyone, from a casual traveler to an airline analyst.

A simple change in date can cut a ticket price by nearly 40%.

[Home](#)[Exploratory Data Analysis](#)[Price Prediction](#)

Price Prediction

Source	Destination
Delhi	Mumbai
Airline	Day of Journey
Air India	9
Month	Duration
March	130 mins
Total Stops	Total Stops (mines (mins)
1	0

Predicted Fare: 10,000 INR

[Home](#)[Exploratory Data Analysis](#)[Price Prediction](#)

Flight Fare Predictor

Source	Destination
Delhi	Mumbai
Airline	Day of Journey
Air India	24
Month	Duration
March	130 mins
Total Stops	Total Stops (mines (mins)
1	0

Predicted Fare: 6,000 INR

By holding all other factors constant, the tool allows users to instantly see the financial impact of a single decision, like choosing a different travel day.

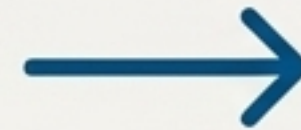
From a complex problem to a deployed, data-driven solution.



Data/Problem



Analysis/Model



Interactive
Application/Solution

This project demonstrates an end-to-end data science workflow. We started with a real-world challenge, used data exploration to uncover deep insights, built and validated a highly accurate predictive model, and delivered the solution as a usable, interactive application. This tool empowers both travelers to save money and airlines to make smarter pricing decisions.